PRE-MIXED CONCRETE

MES-ST-SPC-0004

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Table of Contents

1. Introduction 4

1.1 Scope 4

1.2 Purpose 4

1.3 Conflicts and Deviations 4

2. Key Terms and definitions 4

3. Referenced Documents 5

3.1 Statutory Regulations 5

3.2 Industry Codes and Standards 5

3.2.1 Local Codes and Standards (for Saudi Arabia) 5

3.2.2 International Codes and Standards 6

3.2.3 Alternative Codes 9

3.3 Ma'aden Standards 9

4. Materials 10

4.1 Cement 10

4.1.1 Properties 10

4.1.2 Control, Handling and Storage of Material 11

4.2 Aggregates 11

4.2.1 General 11

4.2.2 Coarse Aggregate 11

4.2.3 Fine Aggregate 11

4.2.4 Control, Handling and Storage of Material 11

4.3 Supplementary Cementitious Material 12

4.4 Water 13

4.5 Additives 13

4.6 Handling, Storage and Batching of Materials 14

5. Concrete Mixes 14

5.1 Proportioning and Mixing 14

5.2 Delivery of Concrete 20

6. Inspection and Testing 20

6.1 Cement 21
6.2 Aggregates 21
6.3 Water 21
6.3.1 General 21
6.3.2 Contained in Aggregates 22
6.4 Admixtures 22
6.5 Trial Mixes 22
6.6 Concrete Quality Control 23

1. INTRODUCTION

1.1 Scope

This document specifies the minimum requirements for supply and delivery of pre-mixed concrete for all Ma'aden projects.

This document is intended to be used in conjunction with the applicable documents listed herein.

1.2 Purpose

The purpose of this specification is to set out the minimum standards for supply of materials, mixing and testing of pre-mixed concrete.

1.3 Conflicts and Deviations

Any deviation from any of the requirements stipulated in this document shall go thru a waiver approval process as per Ma'aden Capital Investment System (MCIS) guidelines and shall be addressed to Ma'aden's Project Development & Engineering (PDE) team for their review and approval.

2. KEY TERMS AND DEFINITIONS

Employer	Saudi Arabian Mining Company (Ma'aden) and its authorized representatives and respective affiliates.
Contractor	Party responsible for supplying materials, plant, labor and supervisory services
Manufacturer	Party who produces and warrants performance of products, materials, and/or items provided in accordance with contract documents
Supplier	Party responsible for supplying concrete materials in accordance with contract documents
GGBFS	Ground Granular Blast Furnace Slag
Sabkha	Sabkha is a saline (generally coastal) deposit consisting of saturated, loose silty sand and possibly clay.

Environmental Engineering Concrete (EEC) Structures	Concrete structures intended for conveying, storing or treating water, wastewater or other non-hazardous liquids and secondary containment of hazardous materials.
Hazardous Material Containment Concrete (HMCC) Structures	Concrete Structures intended for storing hazardous (e.g. ignitable, corrosive, reactive or toxic etc.) chemicals as the primary containment.
Non-Environmental Concrete Structures	Concrete structures other than EEC or HMCC structures

3. REFERENCED DOCUMENTS

This specification requires compliance with the provisions of the latest revision of the relevant standards and codes of the following authorities and technical organizations, as applicable, and specifications which shall be considered to form part of this specification, where applicable. In general, the Contractor shall be responsible for full compliance with applicable Ma'aden Health, Safety and Environmental Standards (MSHEMs). Any deviation from the MSHEMs shall require written approval from the Employer.

3.1 Statutory Regulations

All work performed and materials furnished shall comply with the applicable statutory regulations, codes and other requirements of the Kingdom of Saudi Arabia or local requirements of Project Site. The Contractor shall ensure that he is fully aware and informed of all Government laws, local ordinances, regulations and all orders and decrees of bodies or tribunals having any jurisdiction or authority over the Project and/or Site, which in any manner affect the Project and those engaged or employed on the Project, or which in any way affect the conduct of the Work. The Contractor shall at all times observe and comply with all such Government and/or Ministry laws, bylaws, ordinances, codes, regulations, orders and decrees.

3.2 Industry Codes and Standards

The work shall conform to the following applicable codes and standards.

3.2.1 *Local Codes and Standards (for Saudi Arabia)*

All Ma'aden facilities shall comply with latest version of all Saudi Arabia applicable local codes and standards including but not limited to:

SBC 304	Saudi Building Code – Structural Concrete
SASO/SSA142	Physical or Mechanical Methods of Testing of Portland Cement
SASO/SSA143	Portland Cement Ordinary and Rapid Hardening.

For project locations other than Saudi Arabia, the Contractor shall follow the respective applicable local regulations, codes and standards

3.2.2 *International Codes and Standards*

3.2.2.1. American Concrete Institute (ACI)

ACI SPEC-117	Specification for Tolerances for Concrete Construction and Materials
ACI PRC-201.2	Durable Concrete- Guide
ACI PRC-211.1	Selecting Proportions for Normal-Density and High-Density Concrete-Guide
ACI PRC-211.2	Standard Practice for Selecting Proportions for Structural Lightweight Concrete
ACI PRC-214	Guide to Evaluation of Strength Test Results of Concrete
ACI PRC-232.2	Report on the Use of Fly Ash in Concrete
ACI PRC-234	Guide for the Use of Silica Fume in Concrete
ACI SPEC-301	Specifications for Concrete Construction
ACI PRC-304	Guide for Measuring, Mixing, Transporting and Placing Concrete
ACI PRC-304.2	Guide to Placing Concrete by Pumping Methods
ACI PRC-305	Guide to Hot Weather Concreting
ACI CODE-318RM	Commentary on Building Code Requirements for Reinforced Concrete
ACI PRC-350.2	Hazardous Material Containment in Concrete Structures—Report
ACI 440.1R	Guide for the Design and Construction of Structural Concrete Reinforced with Fiber-Reinforced Polymer (FRP) Bars
ACI PRC-363	Report on High-Strength Concrete

3.2.2.2. American Society for Testing Materials (ASTM)

ASTM C33/C33M	Standard Specification for Concrete Aggregates
ASTM C39/C39M	Standard Test Method for Compressive Strength of Cylindrical Concrete Specimens
ASTM C40/C40M	Standard Test Method for Organic Impurities in Fine Aggregates for Concrete
ASTM C88/C88M	Standard Test Method for Soundness of Aggregates by Use of Sodium Sulfate or Magnesium Sulfate
ASTM C94/C94M	Standard Specification for Ready-Mixed Concrete
ASTM C117	Standard Test Method for Material Finer than 75-μm (No. 200) Sieve in Mineral Aggregates by Washing
ASTM C127	Standard Test Method for Relative Density (Specific Gravity) and Absorption of Coarse Aggregate
ASTM C128	Standard Test Method for Relative Density (Specific Gravity) and Absorption of Fine Aggregate
ASTM C131/C131M	Standard Test Method for Resistance to Degradation of Small-Size Coarse Aggregate by Abrasion and Impact in the Los Angeles Machine
ASTM C136/C136M	Standard Test Method for Sieve Analysis of Fine and Coarse Aggregates
ASTM C138/C138M	Standard Test Method for Density (Unit Weight), Yield and Air Content (Gravimetric) of Concrete
ASTM C142/C142M	Standard Test Method for Clay Lumps and Friable Particles in Aggregates
ASTM C143/C143M	Standard Test Method for Slump of Hydraulic Cement Concrete
ASTM C150/C150M	Standard Specification for Portland Cement
ASTM C172/C172M	Standard Practice for Sampling Freshly Mixed Concrete
ASTM C173/C173M	Standard Test Method for Air Content of Freshly Mixed Concrete by the Volumetric Method

ASTM C183/C183M	Standard Practice for Sampling and the Amount of Testing of Hydraulic Cement
ASTM C186	Standard Test Method for Heat of Hydration of Hydraulic Cement
ASTM C191	Standard Test Method for Time of Setting of Hydraulic Cement by Vicat Needle
ASTM C192/C192M	Standard Practice for Making and Curing Concrete Test Specimens in the Laboratory
ASTM C204	Standard Test Method for Fineness of Hydraulic Cement by Air-Permeability Apparatus
ASTM C231/C231M	Standard Test Method for Air Content of Freshly Mixed Concrete by the Pressure Method
ASTM C260/C260M	Standard Specification for Air-Entraining Admixtures for Concrete
ASTM C457/C457M	Standard Test Method for Microscopical Determination of Parameters of the Air-Void System in Hardened Concrete
ASTM C494/C494M	Standard Specification for Chemical Admixture for Concrete
ASTM C586	Standard Test Method for Potential Alkali Reactivity of Carbonate Rocks as Concrete Aggregates (Rock-Cylinder Method)
ASTM C595/C595M	Standard Specification for Blended Hydraulic Cements
ASTM C618	Standard Specification for Coal Ash and Raw or Calcined Natural Pozzolan for Use in Concrete
ASTM C642	Standard Test Method for Density, Absorption, and Voids in Hardened Concrete
ASTM C685/C685M	Standard Specification for Concrete Made by Volumetric Batching and Continuous Mixing
ASTM C989/C989M	Standard Specification for Slag Cement for Use in Concrete and Mortars
ASTM C1105	Standard Test Method for Length Change of Concrete Due to Alkali-Carbonate Rock Reaction

ASTM C1202	Standard Test Method for Electrical Indication of Concrete's Ability to Resist Chloride Ion Penetration
ASTM C1218	Standard Test Method for Water-Soluble Chloride in Mortar and Concrete
ASTM C1240	Standard Specification of Silica Fume Used in Cementitious Mixtures
ASTM C1293/C1293M	Standard Test Method for Determination of Length Change of Concrete due to Alkali-Silica Reaction
ASTM C1556	Standard Test Method for Determining the Apparent Chloride Diffusion Coefficient of Cementitious Mixtures by Bulk Diffusion
ASTM C1585	Standard Test Method for Measurement of Rate of Absorption of Water by Hydraulic- Cement Concretes

3.2.2.3. British Standards Institution (BSI)

BS 812	Testing Aggregates
BS 1881	Testing Concrete: Methods for analysis of hardened concrete

3.2.2.4. National Ready Mixed Concrete Association (NRMCA)

Checklist for Ready Mixed Concrete Production Facilities.

3.2.3 *Alternative Codes*

Alternative codes may be used for pre-mixed concrete design if approved by the Employer.

3.3 Ma'aden Standards

MD-101-SMEM-EG-ST-SPC-0005	Concrete Works
MD-101-SMEM-EG-ST-SPC-0006	Steel Reinforcing Bars and Wire Fabric
MD-101-SMEM-EG-ST-SPC-0008	Protection Coating for Concrete

4. MATERIALS

4.1 Cement

4.1.1 *Properties*

Portland cement shall conform to ASTM C150/C150M, Type I – for general use when the special properties specified for any other type are not required or Type V – when high sulfate resistance is desired, unless otherwise specified in the Contract Document. Only one brand of cement shall be used.

Cement shall meet the additional requirements specified in Table 4-1.

Table 4-1: Additional Requirements

Properties	Limit for Type I Cement	Limit For Type V Cement
Alkalis ($Na_2O+0.658\ K_2O$)	0.6% maximum	0.6% maximum
Tricalcium Aluminate ($3CaO•Al_2O_3$)	8% – 14%	3% – 5%
Tricalcium Silicate ($3CaO•SiO_2$)	40% – 60%	40% – 60%
Tetracalcium Aluminoferrite ($4CaO•Al_2O_3•Fe_2O_3$)	8% – 14%	15% maximum
Fineness (Air Permeability) in m^2/kg	400 maximum	400 maximum
Time of setting – Vicat Needle, ASTM C191		
Initial set (minutes)	120 minimum	120 minimum
Final set (hours)	6 maximum	6 maximum

In case of conflict with ASTM C150/C150M, the most stringent requirements shall apply, as shown in Table 4-1.

Cement manufactured more than six months before concrete production shall not be used.

Cement shall be used in the order in which it is received at plant.

Different types and brands of cement shall not be mixed together.

Each consignment of cement shall be stocked separately and shall be approved prior to use. The testing and approval shall be conformed to ASTM C150.

4.1.2 *Control, Handling and Storage of Material*

On site, cement shall be placed in weather-proof structures or containers in accordance with ACI SPEC-301, or as per the Manufacturers' recommendation whichever is more stringent.

4.2 Aggregates

4.2.1 *General*

Aggregates shall conform to ASTM C33/C33M and shall be obtained from a single source.

If carbonate or dolomitic materials are proposed, they shall be checked for alkali reactivity in accordance with ASTM C586 or ASTM C1105 or ASTM C1293/C1293M.

4.2.2 *Coarse Aggregate*

Coarse aggregate shall be clean, uncoated, processed aggregate containing no clay, mud, loam or foreign matter and shall be as follows:

- Crushed stone, processed from natural rock or stone
- Washed gravel, either natural or crushed.

Aggregate delivered to the stockpile prior to being washed at quarry shall be washed with potable water and left to dry before charging into the bins. Coarse aggregate shall be selected, re-crushed, finish screened and rinsed as necessary. Coarse aggregate shall comply with the requirements of ASTM C33/C33M, except the magnesium sulfate soundness loss shall be less than 20%.

4.2.3 *Fine Aggregate*

Fine aggregate shall consist of crushed gravel, crushed stone or natural sand with rounded or subrounded particles. Fine aggregates shall be selected to comply with the requirements of ASTM C33/C33M. Natural sand may need to be washed with potable water to achieve these requirements.

4.2.4 *Control, Handling and Storage of Material*

Aggregates shall be stored and handled as specified in ACI SPEC-301.

4.3 Supplementary Cementitious Material

Fly Ash, Silica Fume, Ground Granular Blast Furnace Slag (GGBFS) or Natural Pozzolana may be used as a Supplementary Cementitious provided it can be proven that none of the strength, Chloride resistance or other important properties of the concrete are detrimentally affected.

Silica fume shall be in accordance with ASTM C1240. Fly ash shall be in accordance with ASTM C618, Class F.

The amount of cement replacement shall be established by laboratory tests but shall be between 20 to 30% replacement by weight with fly ash or between 7 to 8% with silica fume.

Blended cement shall meet the requirements for cement and shall conform to all other requirements of ASTM C595/C595M.Natural pozzolan shall be accordance with ASTM C618. GGBFS shall be accordance with the ASTM C989.

When used, supplementary cementitious materials (SCMs) shall comply with the minimum and maximum percent by mass of the cement plus SCMs as indicated in Table 4.3. Additional requirements for SCMs percentages shall be as per ACI 318 or ACI 350.

Table 4.3: Supplementary Cementitious Materials (SCMs)

SCMs	% OF TOTAL CEMENTITIOUS MATERIAL BY MASS	
	MINIMUM	MAXIMUM
Fly Ash – Class F	15	25
Natural Pozzolan	15	25
Silica Fume	5	10
Slag Cement (1)	35	50
Slag Cement (2)	50	50
Total = (Fly ash + Natural Pozzolan) + Silica Fume	(Note-3)	35 [(4) (6)]
Total = (Fly ash + Natural Pozzolan) + Silica Fume + Slag Cement	(Note-3)	50 [(5) (6)]

Notes:

(1) For exposure class other than S1 & S2

(2) For exposure class S1 and S2

(3) The minimum replacement percentages for the individual SCMs may be reduced from those indicated in above table for individual SCM, if the sum of the fractional portions of those percentages provided is greater than or equal to one. For example, when silica fume and slag are used together, the silica fume percentage may be reduced to one-third of the minimum silica fume percentage given in the table, provided that the slag percentage is at least two-thirds of the minimum slag percentage.

(4) Fly Ash and Natural Pozzolan (except Silica Fume) shall be ≤ 25 percent, and Silica Fume shall be ≤ 10 percent, of the total mass of the cementitious materials.

(5) Fly Ash and Natural Pozzolan (except Silica Fume) shall be ≤ 25%, slag cement shall be ≤ 50%, and silica fume shall be ≤ 10% of the total mass of the cementitious materials.

(6) For any combination, total materials of SCMs shall not exceed 50 percent of the total mass of the cementitious materials

4.4 Water

Potable water shall be used in the production, transportation, placement and mixing of concrete, production of ice used in concrete operations, washing aggregates, cooling aggregates, curing of concrete and cleaning forms. If non-potable water is to be used, it shall meet the requirement of Section 3.4 of SBC 304.

4.5 Additives

Water reducing agents, set retarders or other admixtures shall be used in accordance with the Manufacturer's recommendations only with the written approval of the Employer to meet the requirements given in Table 5.1, 5.2, 5.3 & 5.4

Admixtures containing Calcium Chloride shall not be used.

Admixtures shall comply with ASTM C494/C494M and shall be used in accordance with ACI SPEC-301 and the admixture Manufacturer's recommendations.

Where two or more admixtures are proposed to be used simultaneously in the same mix, they will preferably be obtained from the same Manufacturer, and the Contractor shall, in addition to his

obligations under the Contract, give a written guarantee from the Manufacturer(s) of their compatibility to the Employer.

4.6 Handling, Storage and Batching of Materials

The handling, storage and batching of materials shall be in accordance with ACI SPEC-301 and ACI PRC-304.

Batching of materials shall be automatic. The batching equipment shall shut off automatically when the preset quantity has been weighed or measured and shall incorporate an alarm to indicate over-batching of any material.

Equipment shall allow for ready adjustment of added quantities to compensate for variations in weight due to moisture in the aggregates and for changing mix proportions. Weighing equipment shall be of the accuracy specified in ACI SPEC-301.

Admixtures shall be dispensed by suitable automatic dosing equipment.

A printed batch record shall be produced automatically with each batch of concrete. It shall record date, time of batching, batch size, cement weight, weight of the cementitious material (if specified), coarse and fine aggregate weights, water and admixture quantity.

It shall further comply to ACI PRC-304.1, Chapter 3.

5. CONCRETE MIXES

5.1 Proportioning and Mixing

Trial mixes shall be made to produce durable concrete having appropriate workability for the expected use, placement, consolidation and temperatures. The trial mix shall establish proportions for obtaining a dense, impermeable concrete to resist the ingress of damaging salts. The chloride permeability, determined as per ASTM C1202, should not be more than 1000 coulombs.

Structural concrete members shall be assigned an exposure class in accordance with the severity of the anticipated exposure of members as per Tables 5.1 & 5.2.

Based on the exposure classes assigned from Tables 5.1/5.2, concrete mixtures and steel reinforcement type shall comply with the most onerous requirements in Table 5.3.

Unreinforced concrete of no significant structural value such as lean concrete for sub-slabs shall have a minimum 28-day design compressive strength of 17 MPa.

Use of pozzolan or slag or any combination of these products as stated in Table 5.3 are limited to structural concrete members only, excluding plant paving, sidewalks, parking slabs, grade slabs for buildings / shelters reinforced with either welded wire fabric or fiber reinforced polymer rebars or steel / FRP fibers.

Table 5.1 - Chloride Exposure Classes

Exposure Class	Below Ground Exposure		Above Ground Exposure
	Water Soluble Chloride (Cl^-) in Soil, % by Weight	**Water Soluble Chloride (Cl^-) in Ground Water, ppm**	**Environmental and Marine Exposure (to concrete)**
C0	0	0	• Concrete dry or protected from moisture by damp proof membrane • Members fully enclosed within buildings
C1(4)	0	0	• Exposure to moisture or water but not to external source of chlorides • Non-coastal areas
C2(5)	Up to 0.1	Up to 2000	• Exposure to water and external source of chlorides in low concentrations e.g. brackish water etc.
C3(6)	0.1 to 0.5	2000 to 10000	• Exposure to moisture and air-borne chlorides in Coastal areas • Permanently submerged in sea water • Exposure to cyclic wet and dry from water other than seawater (e.g., structures subjected to intermittent brackish wash water)
C4	More than 0.5	More than 10000	• Concrete in tidal and splash zones in marine & offshore environment • Concrete subjected to cyclic wet and dry from seawater or spray from chlorides • Concrete structures containing / handling seawater
	Concrete in contact with Sabkha unless replaced with good soil		

Notes:

(1) Water soluble chloride in ground water need to be identified if water table is less than one meter below the bottom of the foundation or structural element

(2) Water-soluble chloride ion in soil shall be determined in all cases in accordance with AASHTO T291.

(3) ppm: parts per million are measured after soil is compacted.

(4) C1 exposure - Examples: Storm water system and associated catch basins, ponds, culverts

(5) C2 exposure - Examples: Potentially contaminated drainage system, manholes, catch basins, first flush ponds, oil pits, secondary containments for non-hazardous chemicals tanks

(6) C3 exposure - Examples: cooling towers & basins (water other than seawater), storage tanks, pits, ponds & secondary containment for hazardous chemicals

Table 5.2 - Sulfate Exposure Classes

Exposure Class	**Water Soluble Sulfate (SO4) in Soil, % by Mass (ASTM 1580)**	**Dissolved Sulfate (SO4) in water, ppm (ASTM D516 & ASTM D4130)**
S0	SO4 < 0.1	SO4 < 150
S1	0.10 ≤ SO4 < 0.20	150 ≤ SO4 < 1500 or seawater
S2	0.2 ≤ SO4 < 2.00	1500 ≤ SO4 < 10000
S3	SO4 > 2.0	SO4 ≥ 10000

Table 5.3 - Requirements for Reinforced Concrete by Exposures Class (EC)

Exposure Class (EC)	**28 Days Min. f'c (MPa)**	**Cementitious Material Type** [1]	**Minimum Cementitious Content kg/m3** [2]	**Maximum Water to Cementitious material Ratio** [3]	**Maximum water- soluble Cl- ion content in concrete, percent by mass of cementitious materials** [4] [5]		**Reinforcing Steel**	
					Non-Env Str *	EEC Str **	Non-Env Str *	EEC Str **
S0	30	Type I	320	0.45	-	-	Uncoated	Uncoated
S1	30	Type V	340	0.45	-	-	Uncoated	Uncoated
S2	35	Type V	360	0.4	-	-	Uncoated	Uncoated
S3	40	Type V + Pozzolan [6] or slag cement	380	0.4	-	-	Uncoated	Uncoated
C0	30	Type I	320	0.45	1	1	Uncoated	Uncoated
C1	30	Type I	320	0.45	0.3	0.3	Uncoated	Uncoated
C2	30	Type I + Pozzolan [6] or slag cement	340	0.4	0.15	0.1	Uncoated	Uncoated
C3	35	Type I + Pozzolan [6] or slag cement	360	0.4	0.15	0.1	Uncoated [7] or Coated [7]	Coated [8]
C4	40	Type I + Pozzolan [6] or slag cement	380	0.35	0.15	0.1	Coated [9]	CP [10] / Coated [10]

* Non-Env Str - Non-Environmental Concrete Structures. For details refer Definitions.

** EEC Str - Environmental Engineering Concrete structures. For details refer Definitions.

Notes:

(1) For concrete exposed in service to both chlorides and sulfates, the cement type specified for chloride exposure shall be used.

(2) Cementitious contents shall include both cement and supplementary cementitious contents.

(3) For concrete exposed in service to both chlorides and sulfates, the lowest applicable water-cementitious materials shall be used.

(4) Water-soluble chloride ion content that is contributed from the ingredients including water, aggregates, cementitious materials, and admixtures shall be determined in the concrete mixture in accordance with ASTM C1218 at age between 28 and 42 days.

(5) The mass of supplementary cementitious materials (SCMs) used in determining the chloride content shall not exceed the mass of the Portland cement. For proportioning of SCMs, refer Table 4.3.

(6) Pozzolan can be silica fume, fly ash or natural pozzolan. These SCMs shall be tested prior to use on monthly basis or when changing the source of the materials.

(7) EC C3 - For non-environmental concrete structures / foundations use, uncoated rebars with corrosion inhibiting admixtures (non-migratory type) in concrete mix or hot dip galvanized (HDG) rebars or Fusion bonded epoxy coated (FBEC) rebars.

(8) EC C3 - For environmental engineering concrete (ECC) structures & hazardous material containment concrete (HMCC) structures, shrinkage compensating concrete with HDG rebars or FBEC rebars shall be used. Use ACI 350.2, requirements for HMCC structures for the protection of concrete from chemical attack.

(9) EC C4 - For non-environmental concrete structures / foundations, HDG rebars or FBEC rebars shall be used.

(10) EC C4 - For environmental concrete structures, shrinkage compensating concrete, uncoated rebars with Cathodic Protection (CP) system shall be used or shrinkage compensating concrete with HDG rebars shall be used.

(11) In a concrete element, if the reinforcement is provided only for creep / shrinkage, or a minimum percentage and not subjected to significant tensile stresses, the protective

measures mentioned in Notes 7, 8, 9, 10 shall be excluded. e.g. foundations supporting dynamic equipment like pumps, compressors, etc.

(12) On top of the above protection measures mentioned in notes 7, 8, 9, 10, the protective coatings also shall be applied as per Ma'aden standard MD-101-SMEM-EG-ST-SPC-008 – Protective Coatings for Concrete.

Fiber Reinforced polymer Bars (FRP) shall be used as reinforcement for non-critical structural elements. These shall Confirm to ACI 440.1R

The maximum water-Cementitious materials ratio for all concrete shall be as per SBC 304 based on Durability requirements. The weight of the cementitious materials shall include all cementitious material such as fly ash and silica fume.

Classes of concrete shall meet the requirements of Table 5-4.

Table 5-4: Concrete Classes

Class of Concrete	28-Day Compressive Strength (MPa)	Minimum Content of Cementitious Material (kg/m3)
C20	20	300
C30	30	325
C40	40	350
C50	50	375

All classes of concrete may be designated for placing underwater by adding "u" at the end of the Class (i.e. C50u).

Each class of concrete may consist of one or more mixes determined by the maximum size of aggregate and type of admixtures or special aggregates used.

Each mix within a class shall be considered a specific type requiring approval of the design mix.

Unless otherwise specified in the Contract Document, the maximum size of coarse aggregate shall not exceed the sizes given in ACI SPEC-301.

Maximum water-soluble chloride ion concentrations in hardened concrete shall be 0.15% by weight of cement, as indicated in SBC 304, Table 4.4.1, for "reinforced concrete exposed to chloride in service," unless otherwise specified in the Contract Document.

Concrete envelopes for underground electrical ducts and cover slabs for direct-buried cables shall be colored red by adding red oxide powder at the rate of 10 kg/m3 of concrete.

Concrete envelopes for underground instrument air-line ducts shall be colored yellow by adding yellow oxide powder at the rate of 10 kg/m3 of concrete.

Job-mixed concrete shall not be used.

Required durability to comply with ACI PRC-201.1 and NRMCA guides.

5.2 Delivery of Concrete

Delivery equipment shall conform to ACI SPEC-301 and ACI PRC-304 Chapter 4 – Mixing and Transportation. The delivery trucks shall comply with ASTM C94/C94M requirements, shall have NRMCA certification and equipped with a calibrated working resolution, counter as a minimum.

The temperature of low heat concrete shall be within a range of 10°C to 24°C at the point of delivery, unless otherwise agreed with or requested by the Employer in writing. The temperature of all other concrete mixes shall be within a range of 10°C to 32°C at the point of delivery, unless otherwise agreed with or requested by the Employer in writing.

Batch plant mixed concrete shall be transported to the point of delivery either in a truck agitator or a mobile mixer operating at agitating speed.

The total elapsed time between the addition of water to the mix during production and discharging the completed mix at the delivery point shall not exceed 60 minutes unless otherwise approved by the Employer.

The Contractor shall submit identification certificates, with each load of concrete delivered. The following additional information shall be included on the identification certificates:

- The concrete element or part of the works for which the concrete is intended
- No water or other material shall ever be added after the truck leaves the plant
- The amount of water added, if any, at the point of delivery
- Admixture dosage.

6. INSPECTION AND TESTING

The contractor shall comply with the Project Quality Requirements for specific QA/QC requirements

6.1 Cement

Cement shall be tested regularly at monthly intervals at a registered laboratory for compliance with the requirements of ASTM C150/C150M. Test certificates shall be clearly marked with the date of manufacture, source of supply and any other information necessary to identify consignments that are represented by tests and signed by an authorized signatory.

Cement stored for a period in excess of three months from the date of manufacture shall be retested for compliance with ASTM C150/C150M.

6.2 Aggregates

Regular sampling and testing of aggregate materials shall be carried out by a registered testing laboratory and the test results shall be documented for review by the Employer on a continuing basis.

Testing shall be at least at the following intervals:

- Once every 500 tonnes or part thereof of aggregate for compliance with the grading requirements as specified in Clause 4.2 in this Specification
- Once every 6000 tonnes or parts thereof of aggregate for compliance with the Alkali Silica Reaction requirements as specified in Clause 4.2 in this Specification
- Once every 6000 tonnes or part thereof of aggregate for compliance with the following properties:
    - Particle Density
    - Bulk Density
    - Water Absorption
    - Grading
    - Impurities.

6.3 Water

6.3.1 *General*

Three certified copies of test data from an approved laboratory confirming that the water to be used in the manufacture of concrete meets the requirements of ACI SPEC-301, or requirements of section R3.4 of SBC 304 if non-potable water is used, shall be submitted to the Employer for

approval prior to the initial manufacture of any concrete. Further testing of the water shall be carried out at monthly intervals to ensure continued compliance.

6.3.2 *Contained in Aggregates*

The water content of fine aggregates shall be tested twice daily from samples taken as close as possible to the mixer and more often if variability of moisture content is occurring. Coarse aggregates shall also be tested if free water is present due to cooling sprays or any other cause.

The results of water content tests shall be forwarded with each day's batching records to the Employer.

6.4 Admixtures

Specific acceptance testing in accordance with ASTM C494/C494M shall be carried out for each admixture proposed by the Contractor for use in the various concrete grades, prior to the commencement of production.

Applicable exposure categories and classes, and requirements for concrete by exposure class, to be defined per ACI CODE-318 based on site specific environmental exposure defined by Employer. Required durability to comply with ACI PRC-201.1 and NRMCA guides

6.5 Trial Mixes

Trial mixes shall be carried out for each different grade of concrete. Trial mixes shall have a minimum batch size of 2 m3. Testing of trial mixes shall comply with ASTM C39/C39M.

A total of six cylinders per batch shall be tested in accordance with ASTM C39/ C39M for compressive strength. One cylinder shall be tested at three days, two at seven days and two at 28 days. One cylinder shall be kept in reserve for possible future testing.

Two slump tests shall be taken from each batch in accordance with ASTM C143/C143M.

The weight per unit volume of each batch shall be determined in accordance with ASTM C138/C138M. This determination shall be used to adjust batch quantities to produce the ordered amounts of concrete.

All testing of trial mixes shall be carried out by a registered laboratory to demonstrate that the requirements of this specification have been met.

6.6 Concrete Quality Control

The Contractor shall be responsible for concrete sampling and testing in accordance with ACI SPEC-301 as noted in MD-101-SMEM-EG-ST-SPC-0005, Concrete Work Section 5.3 and ACI PRC-311.5.

For each concrete sample taken, the following test shall be conducted in accordance with ACI SPEC-301:

a) Slump Tests
b) Strength Tests.
c) Temperature.
d) Air Content.

Test cylinders set shall be dated and numbered consecutively. Pre-mix concrete that is mixed in plant shall be tested for every 76m3 of concrete produced, as specified in ACI PRC-311.5.

The test cylinders shall not be tested by the Pre-mixed Concrete Supplier. It shall be tested by an Employer approved independent testing laboratory.

Concrete Quality Control test reports shall be provided to the Employer on a weekly basis, unless noted otherwise in the Contract Documents. Data to be included in the test report shall include, but not be limited to, the following:

a) Specified strength.
b) Date of test.
c) Mix Type.
d) 7- and 28-days compressive strength.
e) Slump, air and concrete temperatures at time of test and time of day concrete was batched.
f) Name of inspector making cylinder and cylinder number.

Any test results of concrete strength below the mix design strength shall be reported to the Employer as soon as possible, drawing inferences from seven days results immediately and the batch of concrete corresponding to the sample notified to the Employer.

Pre-mixed concrete Supplier shall hold a current NRMCA Certificate of Conformance for Concrete Production Facilities. Alternatively, certification from an independent testing agent stating conformance with NRMCA Certification of Ready Mixed Concrete Production Facilities is acceptable.

Durability of concrete, shall be confirmed by the following tests:

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(i) Chlorides: Maximum water-soluble chloride ion concentrations in hardened concrete shall conform to Table 5.3.

(ii) Sulfate: The total water-soluble sulfate content in fresh concrete from all sources, expressed as SO4, when tested in accordance with BS 1881, shall not exceed four (4) percent by weight of dry cementitious material.

(iii) Water absorption and volume of permeable voids in the hardened concrete at age 28 to 42 days shall not exceed three (3) and seven (7) percent respectively, as tested in accordance with ASTM C 642.

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